



Role of seafloor production versus continental basalt weathering in Middle to Late Ordovician seawater $^{87}\text{Sr}/^{86}\text{Sr}$ and climate



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ABSTRACT

The global climate of the Ordovician Period (486.9 to 443.1 Ma) is characterized by cooling that culminated in the Hirnantian glaciation. Chemical weathering of Ca- and Mg-bearing silicate minerals and the subsequent trapping of carbon in marine carbonates act as a sink for atmospheric CO_2 on multi-million-year time scales, with basaltic rocks consuming CO_2 at a greater rate than rocks of granitic composition. The oceanic Sr isotope ratio ($^{87}\text{Sr}/^{86}\text{Sr}$) can act as a geochemical proxy for the relative proportion of basaltic versus granitic weathering. Oxygen isotopes ($\delta^{18}\text{O}$) act as a proxy for paleotemperature and ice volume, providing a useful complement to $^{87}\text{Sr}/^{86}\text{Sr}$ in studies of ancient climate. Previous studies have reported stepwise cooling (increasing $\delta^{18}\text{O}$) during the Middle to Late Ordovician. Combined with Sr and C cycle models, this has led to the hypothesis that continental silicate weathering of mafic material drove Ordovician cooling (e.g., the Taconic Orogeny). However, Sr and C cycle models have not accounted for an apparent rise in sea level and seafloor production in the Middle Ordovician (Darriwilian), which would increase the hydrothermal Sr flux as well as degassing along continental volcanic arcs. Furthermore, some Ordovician studies contain temporal uncertainty between $^{87}\text{Sr}/^{86}\text{Sr}$ and $\delta^{18}\text{O}$ curves if they are not based on paired analyses, which can obscure the relationship between silicate weathering and cooling. Here, we present new paired $^{87}\text{Sr}/^{86}\text{Sr}$ and $\delta^{18}\text{O}$ data from conodont apatite and integrate this with both a deterministic (forward) and stochastic (reverse) modeling approach to argue that increased hydrothermal weathering played a role in driving marine $^{87}\text{Sr}/^{86}\text{Sr}$, specifically an inflection occurring in the *Pygoda serra* conodont zone of the mid-Darriwilian Stage ($\sim 460.9 \text{ Ma} \pm 1 \text{ My}$). This $^{87}\text{Sr}/^{86}\text{Sr}$ inflection is accompanied by an increase in $\delta^{18}\text{O}$, consistent with climate cooling. Clarifying the role of seafloor production for marine $^{87}\text{Sr}/^{86}\text{Sr}$ and the implications for Ordovician cooling allows for a more nuanced understanding of the factors that drive multi-million-year shifts in climate.

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1. Introduction

The Ordovician Period (486.9 to 443.1 Ma) is a valuable case study for the various Earth systems that drive and respond to global climate change. The Ordovician climate is characterized by an up to 20°C drop in low-latitude sea surface temperature with smaller-scale changes superimposed (Trotter et al., 2008; Albanesi et al., 2019; Grossman and Joachimski, 2020; Goldberg et al., 2021; Edwards et al., 2021). The extreme warmth of the Early Ordovician ($\sim 42^\circ\text{C}$) was followed by cooling and ultimately widespread

glaciation in the Hirnantian ($\sim 22^\circ\text{C}$) (Trotter et al., 2008; Albanesi et al., 2019; Grossman and Joachimski, 2020; Goldberg et al., 2021; Edwards et al., 2021). This cooling to mild, modern-like low-latitude sea surface temperatures potentially helped drive the Great Ordovician Biodiversification Event (GOBE) and eventually may have led to the first of the “big five” mass extinctions: the end-Ordovician extinction (e.g., Trotter et al., 2008; Rasmussen et al., 2016). Multiple factors could have contributed to this cooling, including decreased volcanic degassing (e.g., McKenzie et al., 2016), the appearance of the first land plants in conjunction with overall increased organic carbon burial (e.g., Lenton et al., 2012), movement of continents through the Intertropical Convergence Zone (ITCZ; Nardin et al., 2011), and increased weathering of calcium-bearing silicates (e.g., Young et al., 2009; Swanson-Hysell and Macdonald, 2017; Macdonald et al., 2019; Conwell et al., 2022). Un-

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understanding the interplay of these systems and their impact on the timing and magnitude of cooling events throughout the Ordovician represents a longstanding issue.

Previous studies have related Ordovician cooling to exposure of young Ca- and Mg-bearing silicates (i.e., basalts) at low latitudes—such as the uplifted island-arc of the Ordovician Taconic Orogeny in Laurentia—that would have increased weathering rates and drawdown of atmospheric CO₂ (e.g., Shields et al., 2003; Young et al., 2009; Swanson-Hysell and Macdonald, 2017; Macdonald et al., 2019; Conwell et al., 2022). Marine ⁸⁷Sr/⁸⁶Sr reflects the combined influence of Sr fluxes and their isotopic ratios from weathering of continental (granitic) crust (including both riverine and groundwater runoff), hydrothermal alteration of seafloor basalt, and the diagenetic alteration of carbonate sediments (e.g., Basu et al., 2001; Bataille et al., 2017; McArthur et al., 2020). ⁸⁷Sr/⁸⁶Sr decreased throughout the Ordovician, with an inflection in the Middle to Late Ordovician (Darriwilian to Sandbian stages) to a greater rate of decrease (e.g., Saltzman et al., 2014; McArthur et al., 2020). Continental basalt weathering was interpreted to drive this ⁸⁷Sr/⁸⁶Sr inflection (e.g., Shields et al., 2003; Young et al., 2009; Swanson-Hysell and Macdonald, 2017), which is supported by a similarly timed shift in neodymium (Nd) isotopes (Swanson-Hysell and Macdonald, 2017; Conwell et al., 2022). However, questions persist about the intensity of basalt versus more continental composition weathering that can be inferred using Sr and Nd isotopes, which affects our understanding of both changes in the ⁸⁷Sr/⁸⁶Sr of the riverine flux and the magnitude of atmospheric CO₂ drawdown.

In addition to uncertainties in continental basalt weathering, even less is known about how submarine basalt weathering may have affected seawater ⁸⁷Sr/⁸⁶Sr. Previous modeling efforts (e.g., Young et al., 2009) held the hydrothermal Sr flux at mid-ocean ridges constant. Similarly, CO₂ degassing in Young et al. (2009) was adjusted to fit the Ordovician paleotemperature curve rather than being based on geologic evidence, such as seafloor production rates (Gaffin, 1987) or associated continental arc activity (e.g., McKenzie et al., 2016; Marcilly et al., 2021). Furthermore, there is increasing awareness that a full picture of climatic transitions in Earth history is only possible with understanding of weathering and degassing in both continental and marine environments (e.g., Coogan and Gillis, 2018; Müller and Dutkiewicz, 2018).

Ordovician paleotemperature trends based on $\delta^{18}\text{O}$ measurements of conodont apatite or brachiopod calcite were reviewed by Grossman and Joachimski (2020) and Goldberg et al. (2021). Although these compilations support the basic conclusions of Trotter et al. (2008) and Albanesi et al. (2019)—extreme warmth in the Early Ordovician followed by long term cooling that leads to the Hirnantian (end-Ordovician) glaciation—the higher-order structure is subject to debate. Grossman and Joachimski (2020) show an inflection in the $\delta^{18}\text{O}$ curve toward cooling in the late Darriwilian to early Sandbian, approximately coeval with the inflection in ⁸⁷Sr/⁸⁶Sr (Saltzman et al., 2014) and Nd isotopes (Swanson-Hysell and Macdonald, 2017; Conwell et al., 2022), which was not apparent in the coarser resolution $\delta^{18}\text{O}$ curve of Trotter et al. (2008). A late Darriwilian to early Sandbian cooling is further supported by conodont apatite $\delta^{18}\text{O}$ from Männik et al. (2021) in Baltoscandia and Edwards et al. (2021) in North America, although further work is needed from other well-preserved sections throughout the world to demonstrate the significance for global Ordovician climate.

We present new paired ⁸⁷Sr/⁸⁶Sr and $\delta^{18}\text{O}$ data measured on conodont apatite from the Arbuckle Mountains of Oklahoma, which represents one of the best preserved and most well-constrained conodont biostratigraphic data sets for the Darriwilian to Sandbian (~465 to 455 Ma) study interval (Bauer, 2010). We use the eustatic sea level curve of the Middle to Late Ordovician (second order; Haq and Schutter, 2008) to drive changes in hydrothermal Sr flux (e.g., Berner and Kothavala, 2001) using the

isotope mass balance approach of Ingram et al. (1994). The observed link between the ⁸⁷Sr/⁸⁶Sr inflection and the 2nd order eustatic rise at the base of the Tippecanoe megasequence suggests that seafloor production rates may have increased substantially during this time (Shields et al., 2003; Saltzman et al., 2014). Notably, the studied Arbuckle Mountains stratigraphic succession is the global reference standard for this time period in the eustatic sea level curve of Haq and Schutter (2008). We find that increased seafloor production may have contributed to the inflection in seawater ⁸⁷Sr/⁸⁶Sr to lower values but cannot account for the entire Late Ordovician drop, confirming a role for basaltic weathering on land (e.g., Taconic Orogeny; Young et al., 2009; Swanson-Hysell and Macdonald, 2017) in both the ⁸⁷Sr/⁸⁶Sr and Ordovician cooling trend suggested by conodont apatite $\delta^{18}\text{O}$.

2. Background

2.1. Drivers of marine ⁸⁷Sr/⁸⁶Sr in Sr cycle models

The oceanic residence time (~10⁶ years) of Sr is significantly longer than the ocean mixing time (~10³ years), resulting in a global, homogenous ⁸⁷Sr/⁸⁶Sr marine signal (e.g., McArthur et al., 2020). This signal is primarily driven by two Sr fluxes: hydrothermal alteration of seafloor basalt ($15.7 \pm 2.4 \times 10^9$ mol/yr; ⁸⁷Sr/⁸⁶Sr ~ 0.7035) and continental weathering (33.3×10^9 mol/yr; ⁸⁷Sr/⁸⁶Sr ~ 0.7119), which encompasses riverine runoff and—to a smaller extent—groundwater (Elderfield and Schultz, 1996; Basu et al., 2001). Diagenetic Sr from the alteration of deep-sea sediments represents a third flux, (3.4×10^9 mol/yr; ⁸⁷Sr/⁸⁶Sr ~ 0.7084) though it is small enough to not typically be considered a driver of marine ⁸⁷Sr/⁸⁶Sr (Elderfield and Schultz, 1996).

Although the riverine flux is often viewed as the primary driver of seawater ⁸⁷Sr/⁸⁶Sr based on its larger magnitude, the seafloor hydrothermal flux can also play a major role (e.g., Ingram et al., 1994; Jones and Jenkyns, 2001; Kristall et al., 2017; Nana Yobo et al., 2021). The rate of fluid-rock reaction between seawater and basalt at mid-ocean ridges and off-axis ridge flanks is predicated on multiple interrelated factors including magma/heat supply, sedimentation rate, heterogeneity of the aquifer's permeability, ocean bottom water temperature, and tectonic faulting creating channelized flow, all of which affect water-to-rock ratio (e.g., Coogan and Gillis, 2018). A high rate of seafloor production—either in the form of increased spreading rate at mid-ocean ridges or increased length of ridges—influences several of these factors and results in an increased hydrothermal flow, which is directly tied to increased fluid-rock reaction (e.g., Coogan and Gillis, 2018). Therefore, increased seafloor production has been considered a viable driver of decreasing marine ⁸⁷Sr/⁸⁶Sr in modeling studies (e.g., Ingram et al., 1994; Jones and Jenkyns, 2001; Kristall et al., 2017; Nana Yobo et al., 2021). An additional process that may drive changes in the hydrothermal Sr flux is the Ca, Mg, Sr, and SO₄ concentrations of seawater entering mid-ocean ridge (MOR) systems. A shift in the relative amounts of these elements may drive changes in the ⁸⁷Sr/⁸⁶Sr composition of output hydrothermal fluids, such that Ordovician hydrothermal fluid may have been ~0.003 more radiogenic than modern fluids (Antonelli et al., 2017).

2.2. Seafloor spreading and CO₂ degassing in long term climate

Increased seafloor production contributes to increased degassing at both the spreading center and continental volcanic arcs. The latter acts as a more significant source of CO₂ via increased subduction and volcanic activity (e.g., Van Der Meer et al., 2014; McKenzie et al., 2016; Müller and Dutkiewicz, 2018). However, studies such as Jones and Jenkyns (2001) suggest that increased

rates of seafloor production, associated with shifts to lower seawater $^{87}\text{Sr}/^{86}\text{Sr}$, could drive higher CO_2 levels associated with volcanic degassing at spreading centers, in this case during the Jurassic and Cretaceous. Because long-term variations in sea level can reflect changes in seafloor production (e.g., Gaffin, 1987), global carbon cycle models have used sea level to drive degassing (e.g., GEOCARB, in Berner and Kothavala, 2001; COPSE, in Mills et al., 2017).

Reconstruction of the length of subduction zones via full-plate reconstruction modeling (e.g., Merdith et al., 2021) serves as an alternative to estimating seafloor spreading using sea level inversion. However, knowledge of subduction zone length relies on geodynamic modeling (e.g., Van Der Meer et al., 2014), which can be difficult in the Paleozoic without preserved seafloor. An additional method for reconstructing degassing rates at continental volcanic arcs is outlined in McKenzie et al. (2016), using young detrital zircon abundances. Although this represents a novel and promising new approach to estimating degassing, the resolution using detrital zircon abundances is still limited (i.e., $\sim > 10^7$ years).

3. Methods

3.1. Sample location

Samples for this study were collected from a ~ 600 m interval of the Simpson Group exposed along Interstate 35 N in Oklahoma, from the upper McLish, Tulip Creek, and lower Bromide Formations (Fig. 1; Derby et al., 1991; Saltzman et al., 2014). The depositional record of the middle to upper Simpson Group is characterized by successive fluctuations of sea level during a period of overall transgression in the generally shallow-marine setting of the Oklahoma Basin (Derby et al., 1991).

3.2. $^{87}\text{Sr}/^{86}\text{Sr}$ analysis

Ordovician-age rock samples from the Simpson Group were originally collected for conodont microfossils by Bauer (1989). Conodont elements from this section have a low conodont alteration index (CAI ~ 1) and include an even mix of morphologies, including both coniform (cones) and ramiform (platform) elements. Approximately 0.2 mg of conodont elements (based on expected Sr concentration of 5000 to 9000 ppm; Saltzman et al., 2014) were selected for each sample. Conodont elements were sonicated in Milli-Q deionized water to remove debris/contamination. Conodont elements were then dissolved in 6N HCl and the resulting solution analyzed for Sr abundance on an Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES) in the Ohio State University Trace Element Research Laboratory. Sr was separated from the solutions using Eichrom Industries Sr resin in 125 μl Teflon microcolumns, following the method outlined in Scher et al. (2014). Samples were analyzed for $^{87}\text{Sr}/^{86}\text{Sr}$ on a Triton Thermal Ionization Mass Spectrometer (TIMS) at the Ohio State University Thermal Ionization Mass Spectrometry Laboratory. We used SRM 987 as a standard, with an average measured value of 0.710251 (2SD 2×10^{-5} ; $n = 96$; analyzed from 1/31/18 to 3/26/19). We corrected standard $^{87}\text{Sr}/^{86}\text{Sr}$ values to 0.710248 (McArthur et al., 2020) in order to compare with prior results and remove any drift. We accounted for potential ^{87}Rb contamination by measuring ^{85}Rb and using the assumed $^{87}\text{Rb}/^{85}\text{Rb}$ value 0.386000 (Steiger and Jäger, 1977). We accounted for instrument mass fractionation by normalizing $^{87}\text{Sr}/^{86}\text{Sr}$ values to an assumed $^{88}\text{Sr}/^{86}\text{Sr}$ ratio of 8.375 (Steiger and Jäger, 1977).

The uncertainty associated with each individual $^{87}\text{Sr}/^{86}\text{Sr}$ analysis (i.e., internal analytical uncertainty) is defined as the absolute standard error of 200 measurements per sample. On average, this uncertainty is 3×10^{-6} ($n = 33$). The uncertainty associated with replicate sample analyses (i.e., external analytical uncertainty) is

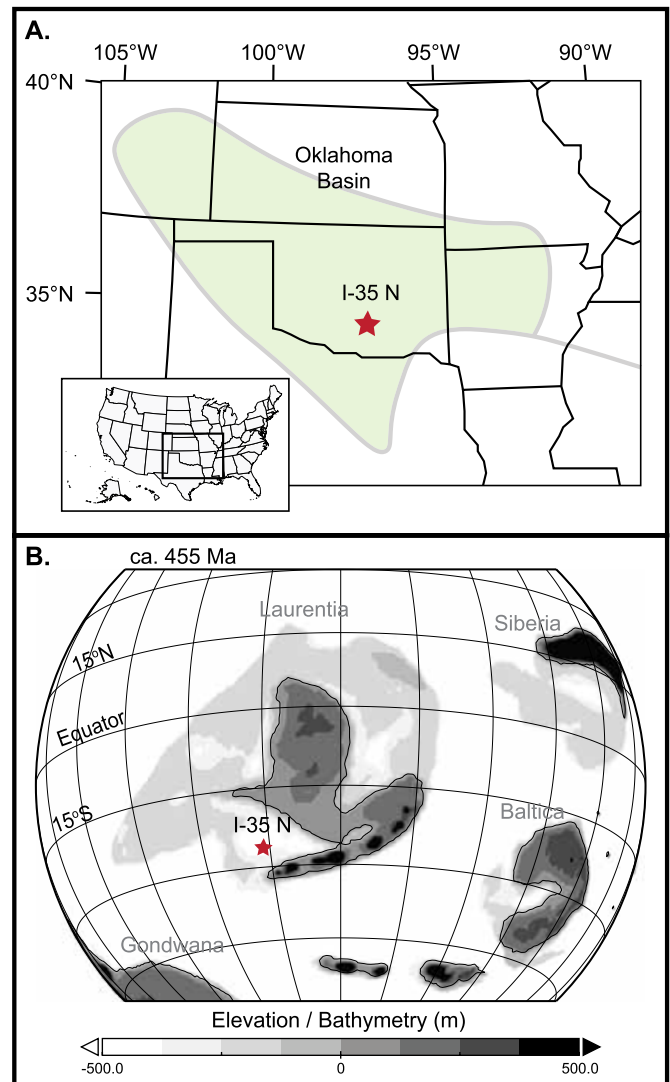


Fig. 1. A) Location of I-35 N site in Arbuckle Mts. Modified from Carlucci et al. (2014). B) Paleogeographic reconstruction for the Late Ordovician and location of I-35 N study site on Laurentia (Scotese and Wright, 2018).

defined as two standard deviations of repeated measurements of standard SRM 987; this value was 8×10^{-6} ($n = 96$). In addition, five samples (83JD-80, -88, -115, -125, -130) were analyzed twice on the same wheel. The average difference between each set of replications was 6×10^{-6} ($n = 5$), while the average (pooled) standard deviation of each set was 9×10^{-6} . Data from the I-35 N site reported by Saltzman et al. (2014) were measured with an average internal and external analytical uncertainty of 9×10^{-6} ($n = 30$) and of 3×10^{-5} ($n = 80$), respectively. These raw data from Saltzman et al. (2014) have been recalibrated for this study using a standard value of 0.710248 for SRM 987 (McArthur et al., 2020). Additional analytical details can be found in Appendix B in Supplementary Material.

3.3. Leaching of conodont samples

To mitigate the potential impact of diagenesis, we completed a set of leaching experiments. Leaching is aimed at removing outer layers of a conodont microfossil that could have been altered, affecting the primary $^{87}\text{Sr}/^{86}\text{Sr}$ value (e.g., Holmden et al., 1996) (see Appendix A in Supplementary Material for additional details). Of the 18 samples analyzed for $^{87}\text{Sr}/^{86}\text{Sr}$, several were run in dupli-

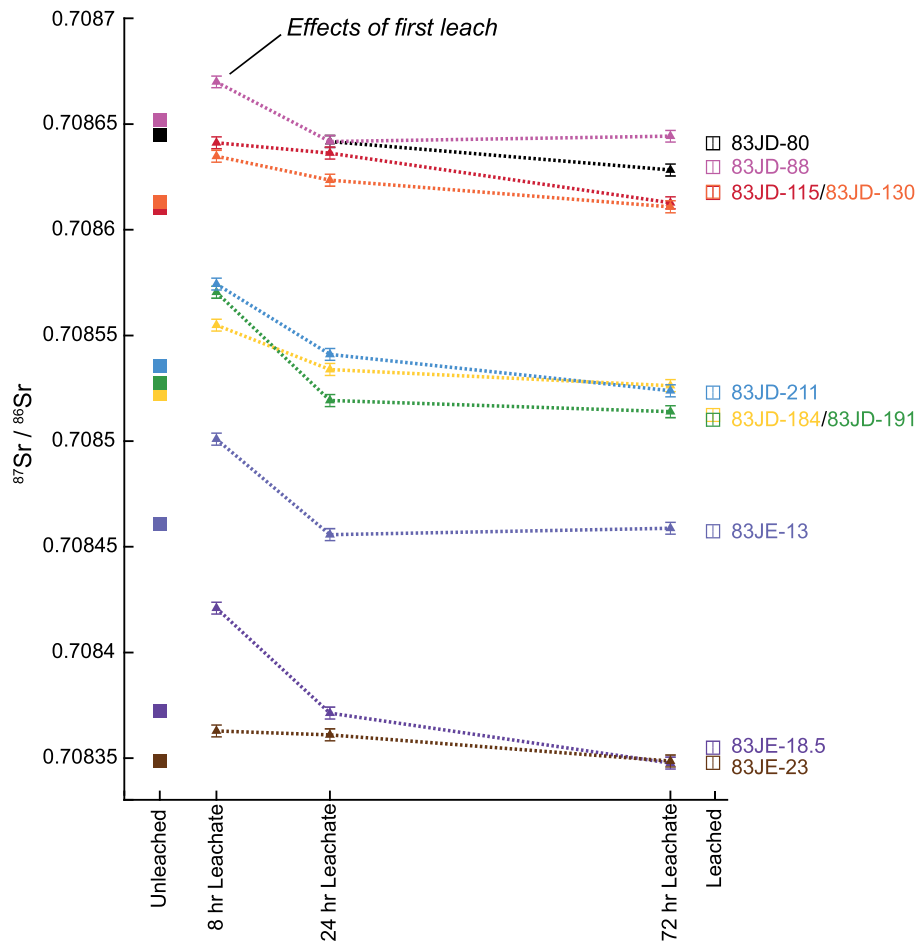


Fig. 2. Ten Ordovician conodont samples from I-35 N were split, with one portion left unleached and the other leached. Leached samples were left in 5% acetic acid for 72 hours with leachate collected at 8, 24, and 72 hrs. Triangles represent leachates at 8, 24, and 72 hrs. The 8-hr leachate from 83JD-80 (black) did not produce useable results. Error bars represent absolute standard error of measurement (internal analytical uncertainty). See Table A.3 in Supplementary Material for details. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

cate or triplicate (Table A.1 in Supplementary Material), and for 10 of those samples, one of the duplicates or triplicates included leaching (designated with an L in Table A.1) in which a subsample of ~ 2 mg of conodont elements was treated for 72 hours in 5% acetic acid. Leached conodont $^{87}\text{Sr}/^{86}\text{Sr}$ values are on average 7×10^{-6} (2SD 2×10^{-5} , $n = 10$) less radiogenic than the corresponding unleached values (Table A.3 in Supplementary Material). A paired t-test suggests this difference is statistically significant with 95% certainty. For two samples (83JD-115 and 83JD-130), the leached fraction is more radiogenic than the unleached fraction (by 7×10^{-6} and 5×10^{-6} respectively).

In addition, the acetic acid used to leach samples (i.e., the leachate) was also analyzed for $^{87}\text{Sr}/^{86}\text{Sr}$ to determine the isotopic value of the Sr being removed from the conodont elements. After 8, 24, and 72 hours, the leachate was collected for analysis (Fig. 2; Table A.3 in Supplementary Material) and fresh 5% acetic acid was added. All leached samples and their leachates were also analyzed for Sr abundance (μg) in order to determine mass balance (Table A.4 in Supplementary Material), i.e., the amount of Sr in each fraction.

The $^{87}\text{Sr}/^{86}\text{Sr}$ value of leachates confirms the results of Holmden et al. (1996). In most samples, the 8-hr leachates are more radiogenic than the other leachates, unleached, as well as leached samples (Fig. 2). The 24-hr leachates are generally more radiogenic than the unleached samples, though often within analytical error. Further, the 72-hr leachates have values comparable to the leached samples. This suggests that the majority of Sr leached from the

outermost layers of conodont elements during the first 8 hours is likely diagenetic in origin. However, for these conodont elements, the diagenetic Sr is a comparatively small portion of the total Sr in a sample, at most 15% but on average 8% of the total Sr (Table A.4 in Supplementary Material). In summary, leaching leads to a lower $^{87}\text{Sr}/^{86}\text{Sr}$ value (on average by 7×10^{-6}). By comparison, external analytical uncertainty is estimated as 8×10^{-6} . As a result, we suggest that all $^{87}\text{Sr}/^{86}\text{Sr}$ values, both leached and unleached, are reliable (see Appendix A in Supplementary Material for more details).

3.4. $\delta^{18}\text{O}$ analysis

Twenty conodont samples from this study and Saltzman et al. (2014) were analyzed for $\delta^{18}\text{O}$ at the University of Erlangen-Nuremberg/Germany (Table A.2 in Supplementary Material). Chemical conversion of apatite bound phosphate into trisilverphosphate (Ag_3PO_4) was performed according to the method described in Joachimski et al. (2009). Oxygen isotope analyses were performed using a TC-EA (high-temperature conversion-elemental analyzer) coupled online to a ThermoFinnigan Delta V Plus mass spectrometer. 0.2 to 0.3 mg Ag_3PO_4 was weighed into silver foil and transferred to the sample carousel of the TC-EA. At 1450°C , the silver phosphate is reduced and CO forms as the analyte gas (Vennemann et al., 2002). CO was transferred in a helium stream through a gas chromatograph via a ConFlo IV interface to the mass spectrometer. All values are reported in ‰ relative to VSMOW. Samples as well

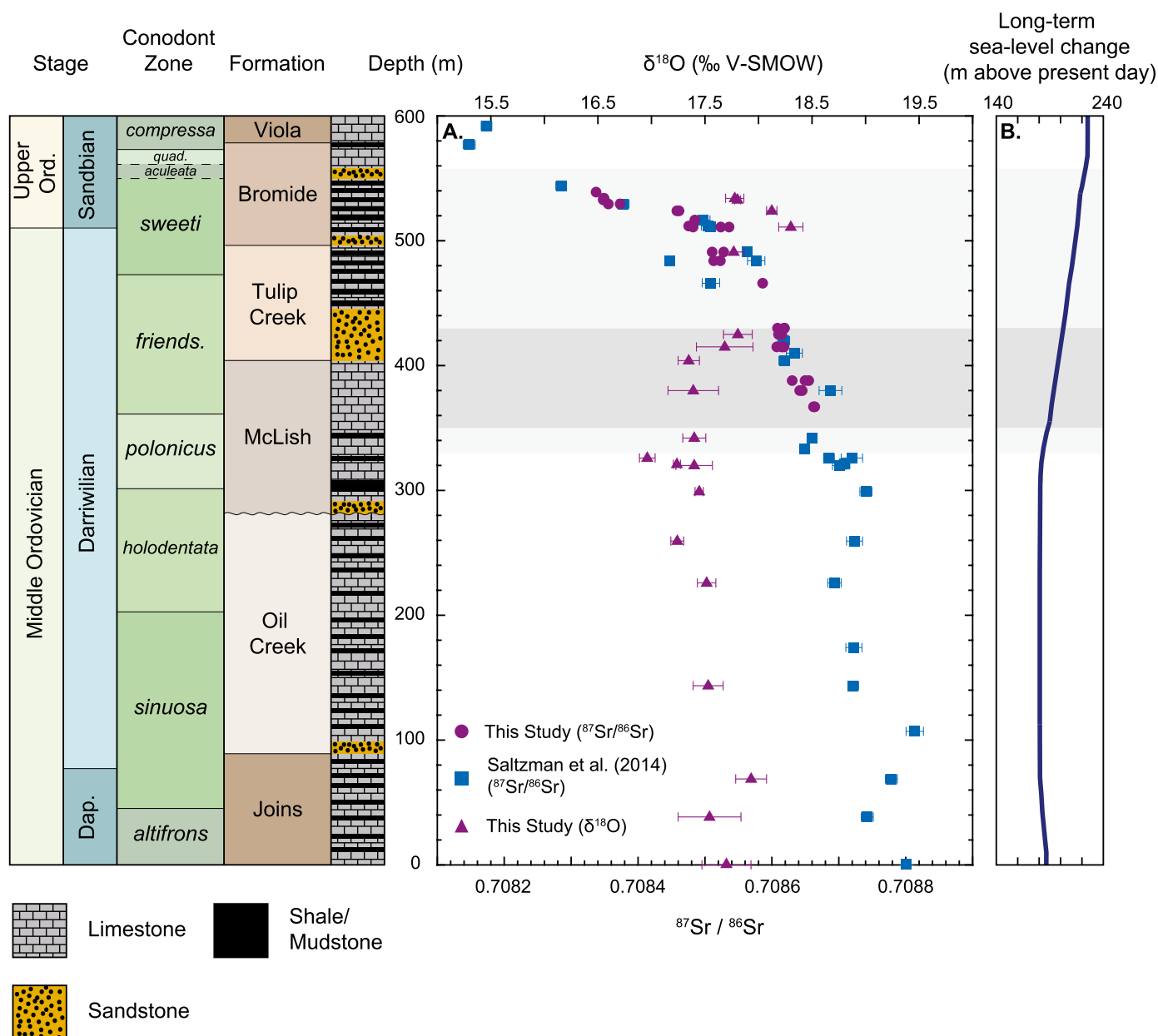


Fig. 3. A) Arbuckle Mountains, Oklahoma, I-35 N section data from this study and Saltzman et al. (2014) plotted over depth. $^{87}\text{Sr}/^{86}\text{Sr}$ error bars represent standard error of measurement (internal analytical uncertainty). $\delta^{18}\text{O}$ error bars represent 1 SD of triplicate sample measurements ($\pm 0.3\text{‰}$). B) Long-term sea level relative to modern day; data from Haq and Schutter (2008).

Possible age range of inflection based on regression analyses outlined in light gray; 95% confidence age range of inflection outlined in dark gray. Conodont zonation derived from Bauer (1989, 2010). Lithology after Bauer (2010) and Carlucci et al. (2014).

as standards were measured in triplicate. The measurements were calibrated by performing a two-point calibration using NBS 120c (21.7‰) and a commercial Ag_3PO_4 (9.9‰). A laboratory standard was used as a control standard and processed together with the samples. All standards were calibrated to TU1 (21.11‰) and TU2 (5.45‰; Vennemann et al., 2002). The uncertainty associated with each individual analysis (i.e., internal analytical uncertainty) is defined as the standard deviation of triplicate sample measurements. On average this value is $\pm 0.13\text{‰}$ ($n = 20$).

4. Results

4.1. $^{87}\text{Sr}/^{86}\text{Sr}$

Eighteen conodont samples from the I-35 N section of the Simpson Group were analyzed for $^{87}\text{Sr}/^{86}\text{Sr}$, with several mea-

sured in duplicate or triplicate (Table A.1 in Supplementary Material; Fig. 3). The average Sr concentration is 7250 ppm (2SD = 2937 ppm; $n = 33$). We combined these new data with conodont-based $^{87}\text{Sr}/^{86}\text{Sr}$ data from Saltzman et al. (2014) and Edwards et al. (2015), resulting in a compilation of 237 North American data points from throughout the Ordovician (Fig. 4). To clarify the numerical age of the $^{87}\text{Sr}/^{86}\text{Sr}$ inflection, we performed 30 regression analyses on the 237 data points (see section B.7 in Supplementary Material). Each analysis produced a possible inflection age, ranging from 463.3 Ma at the earliest to 456.4 Ma at the latest (Table A.5 in Supplementary Material). We averaged the 30 inflection ages and calculated an inflection age range of $460.9 \text{ Ma} \pm 1 \text{ My}$ (95% confidence interval; Fig. 3, 4). This places the $^{87}\text{Sr}/^{86}\text{Sr}$ inflection in the late Darriwilian (upper McLish and lower Tulip Creek Formations), in the upper *Phragmodus polonicus* and lower *Cahabag-*

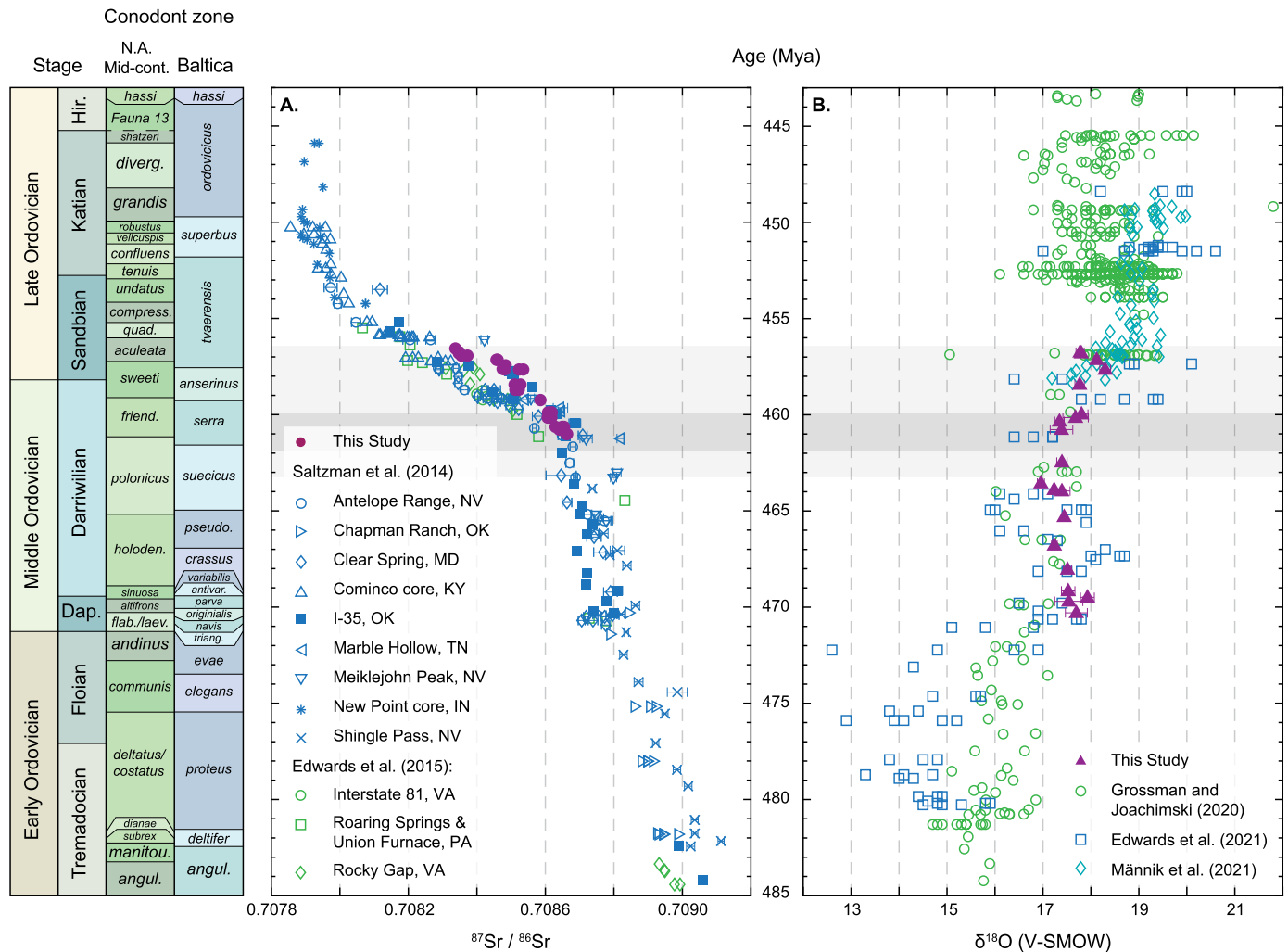


Fig. 4. A) $^{87}\text{Sr}/^{86}\text{Sr}$ data from this study, Saltzman et al. (2014), and Edwards et al. (2015) plotted against age. Error bars represent standard error of measurement (internal analytical uncertainty). B) $\delta^{18}\text{O}$ data from this study, Grossman and Joachimski (2020), Edwards et al. (2021), and Männik et al. (2021) plotted against age. Error bars represent 1 SD of triplicate measurements ($\pm < 0.3\text{‰}$). In general, cooler temperatures correspond to heavier $\delta^{18}\text{O}$ values. Possible age range of inflection based on regression analyses outlined in light gray; 95% confidence age range of inflection outlined in dark gray. Conodont zonation derived from Goldman et al. (2020).

nathus friendsvillensis conodont zones (North American Midcontinent zonation), or almost entirely in the *P. serra* conodont Zone (Baltica zonation; Goldman et al., 2020).

4.2. Conodont apatite $\delta^{18}\text{O}$

Twenty conodont samples from Oklahoma (including samples used in both this study and from Saltzman et al., 2014) were analyzed for oxygen isotopes (Table A.2 in Supplementary Material). $\delta^{18}\text{O}$ shows values of $\sim 17.5\text{‰}$ until ~ 461 to 460 Ma. At this point, $\delta^{18}\text{O}$ starts to shift to higher values reaching $\sim 18.3\text{‰}$ at ~ 458 Ma in the early Sandbian. The $\delta^{18}\text{O}$ shift roughly correlates with the $^{87}\text{Sr}/^{86}\text{Sr}$ inflection in the *C. friendsvillensis*/*P. polonicus* and *P. serra* conodont zones (Fig. 4).

5. Discussion

5.1. Diagenesis and preservation of primary seawater values for $^{87}\text{Sr}/^{86}\text{Sr}$ and $\delta^{18}\text{O}$

Both $\delta^{18}\text{O}$ and $^{87}\text{Sr}/^{86}\text{Sr}$ of conodont apatite can be diagenetically altered, either via recrystallization or exchange with porewaters (e.g., Joachimski et al., 2009; Saltzman et al., 2014). These

processes occur more readily under high-temperature conditions, which darken conodont elements (i.e., high conodont alteration index, or CAI; e.g., McArthur et al., 2020). Conodonts analyzed in this study have a low CAI (~ 1) indicating low thermal overprint (Saltzman et al., 2014).

In the case of $\delta^{18}\text{O}$, phosphate-bound oxygen in conodont apatite is generally considered a robust source of primary marine values (e.g., Trotter et al., 2008; Joachimski et al., 2009; Albanesi et al., 2019; Grossman and Joachimski, 2020). We have compared $\delta^{18}\text{O}$ values from this study with compiled Ordovician $\delta^{18}\text{O}$ data by Grossman and Joachimski (2020) as well as recently published data of Edwards et al. (2021) and Männik et al. (2021; Fig. 4). The data show a slight decrease through the Dapingian and early Darriwilian followed by a shift to higher values (cooling) from the late Darriwilian into the early Sandbian (Fig. 4B). This trend is consistent with data reported from various sites and bolsters confidence in the robust nature of the $\delta^{18}\text{O}$ trend preserved in the Arbuckle Mountain samples.

For $^{87}\text{Sr}/^{86}\text{Sr}$, conodont apatite may compare well with brachiopod calcite as a source of reliable data (e.g., Shields et al., 2003; Saltzman et al., 2014; McArthur et al., 2020). Factors thought to affect seawater $^{87}\text{Sr}/^{86}\text{Sr}$ preservation in conodont apatite include the host rock of the samples from which conodonts are extracted

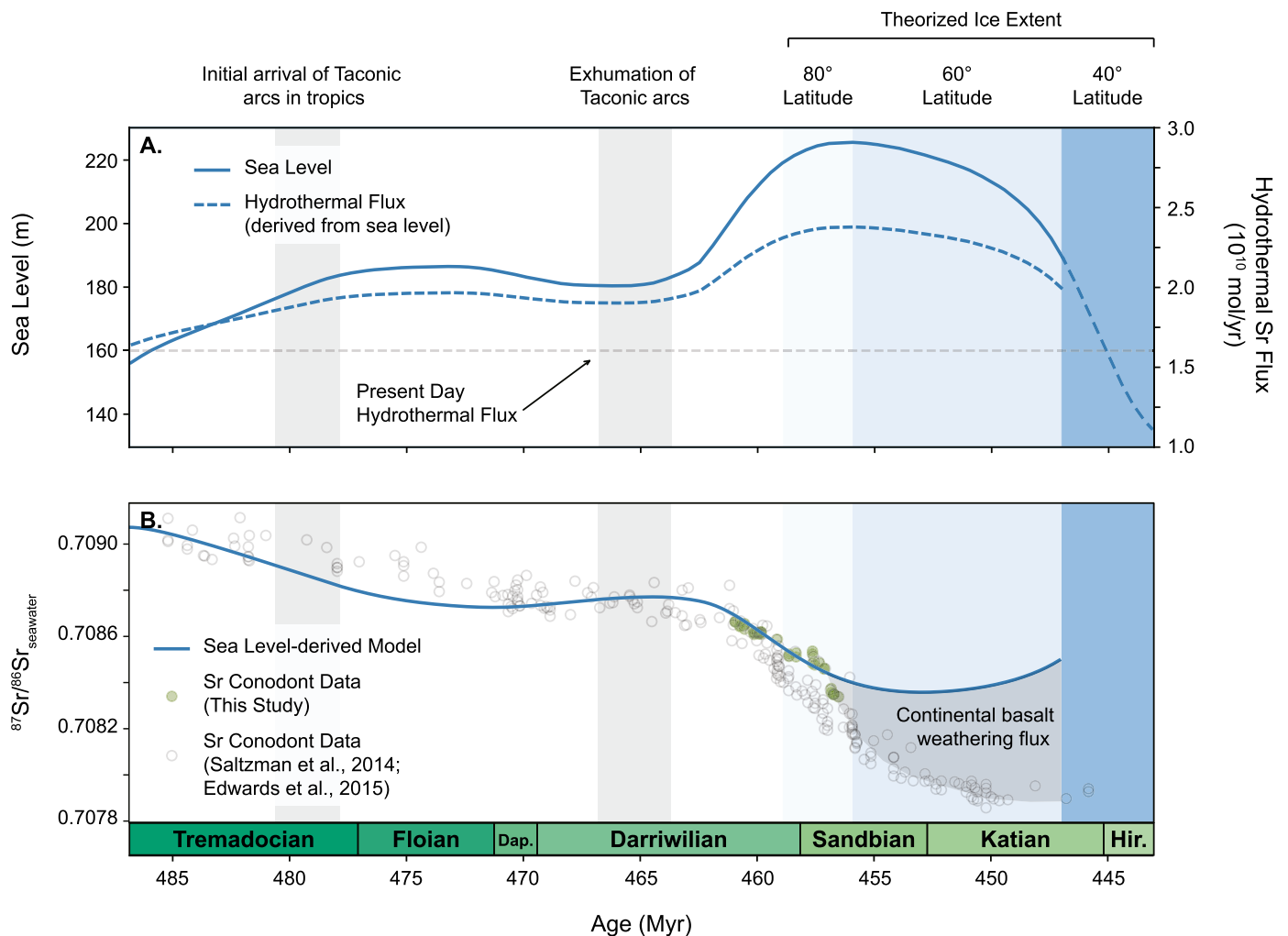


Fig. 5. **A)** Sea level after Haq and Schutter (2008; left axis) and estimated hydrothermal flux (right axis). Present day hydrothermal flux of 1.6×10^{10} (Elderfield and Schultz, 1996) included for reference. **B)** Seawater $^{87}\text{Sr}/^{86}\text{Sr}$ curve resulting from model in which sea level-derived hydrothermal flux and all other factors have been held constant. Sr conodont data from this study and Saltzman et al. (2014). Gap between model and data is likely a result of weathering from continental basalt (i.e., the Taconic Orogeny) and is addressed in stochastic model (see Fig. 6). Gray bars indicate events of Taconic Orogeny; taken from Swanson-Hysell and Macdonald (2017). Blue boxes indicate glacial activity; taken from Goldberg et al. (2021).

(i.e., shale or limestone), which can affect pore-water chemistry and post-mortem Sr exchange, thermal alteration, and the morphology of the conodont elements analyzed (Holmden et al., 1996). We used leaching (see section 3.3) to mitigate the effects of secondary (radiogenic) Sr that may adhere to the conodont surfaces. Similar to $\delta^{18}\text{O}$, we compared our $^{87}\text{Sr}/^{86}\text{Sr}$ results to those of Shields et al. (2003) and Saltzman et al. (2014) and found similar values and trends (Fig. 4A) that increase confidence in preservation of primary seawater values.

5.2. $^{87}\text{Sr}/^{86}\text{Sr}$, $\delta^{18}\text{O}$, and eustatic sea level rise (Tippicanoe megasequence transgression)

The relative timing of changes in $^{87}\text{Sr}/^{86}\text{Sr}$, $\delta^{18}\text{O}$, and sea level can be used to infer causal relationships. We first consider how timing of the $^{87}\text{Sr}/^{86}\text{Sr}$ inflection compares to a major shift in eustatic sea level and the possible implications for rates of seafloor spreading. In the I-35 N section of the Arbuckle Mountains, which is the reference section for the Ordovician portion of the eustatic sea level curve of Haq and Schutter (2008), the $^{87}\text{Sr}/^{86}\text{Sr}$ inflection is recorded from the upper McLish Fm. to the lower Tulip Creek Fm. in the *P. polonicus* and *C. friendsvillensis* North American Midcontinent conodont biozones (Fig. 3, 4). The McLish-Tulip Creek transition—considered conformable based on conodont

chronozones (Derby et al., 1991)—records the onset of the Tippicanoe transgressive sequence (2nd order sea level rise) following the first major megasequence (Sauk-Tippicanoe) boundary of the Paleozoic (Fig. 3). Although the initial transgression may have been as much as one conodont zone earlier in the Oil Creek-McLish formation transition (Fig. 3; Derby et al., 1991), the $^{87}\text{Sr}/^{86}\text{Sr}$ inflection correlates with the most rapid rise in sea level at ~461.5 Ma (Fig. 5).

$\delta^{18}\text{O}$ values also show a small positive trend beginning in the Tulip Creek Formation at ~460 Ma, which is consistent with a shift to cooler temperatures during the late Darriwilian to Sandbian that is seen in the recently published paleotemperature curves for Baltoscandia (Männik et al., 2021) and North American (Nevada; Edwards et al., 2021) (Fig. 4). Männik et al. (2021) refer to this as the 'Late Kulkuse Cooling Event', and although these authors correlate this to a 3rd order eustatic fall in the Haq and Schutter (2008) sea level curve, the long term (2nd order) sea level was rising at this time (Fig. 3).

It is unlikely that glacial melting drove the Middle Ordovician 2nd order rise in sea level (Tippicanoe transgression; Haq and Schutter, 2008). The establishment of Middle Ordovician ice sheets has been inferred (e.g., Rasmussen et al., 2016), although there is not yet unambiguous evidence for glacial deposits (e.g., Cocks and Torsvik, 2020). Glacio-eustasy may be more likely to drive

the higher frequency 3rd order sea level fluctuations (e.g., Männik et al., 2021). The best explanation for the 2nd order rise in sea level during the Tippecanoe transgression (Haq and Schutter, 2008; Fig. 3) is a higher rate of spreading at mid-ocean ridges or a lengthening of ridges that created more young, warm, and buoyant mid-ocean ridge basalts (MORBs) that volumetrically replace seawater (e.g., Gaffin, 1987).

We acknowledge that previous studies such as Marcilly et al. (2021) have used alternative methods—full-plate models and detrital zircons—that produce a *decreasing* seafloor production rate during the Middle to Late Ordovician, which results in hydrothermal Sr fluxes that remain relatively constant during this time period (Fig. A.7 in Supplementary Material). We have chosen not to utilize these hydrothermal fluxes in our models because they focus on much longer time intervals (Phanerozoic scale). They therefore have large bin sizes that result in “smoothing out” of nuances in the seafloor production curve that we believe are best resolved utilizing the approach we have taken with sea-level inversion. If in fact Marcilly et al. (2021) and GEOCARB (Berner and Kothavala, 2001) were the more accurate models, it would suggest that the increase in eustatic sea level is more likely a result of ice sheet melting rather than increased seafloor production as proposed in our study, which in turn would suggest that even greater amounts of continental rather than seafloor basalt weathering are required to produce the Ordovician seawater Sr drop as discussed in the next section.

5.3. Role of hydrothermal weathering on Ordovician seawater $^{87}\text{Sr}/^{86}\text{Sr}$

If, as discussed above, we assume that 2nd order eustatic rise (Tippecanoe transgression) during the Middle to Late Ordovician (Darriwilian to Sandbian Stages) was driven by seafloor production, we can use a Sr isotope mass balance model to test whether changes in the hydrothermal Sr flux can feasibly reproduce the Ordovician $^{87}\text{Sr}/^{86}\text{Sr}$ curve. With the caveat that non-uniqueness affects the interpretation of seawater $^{87}\text{Sr}/^{86}\text{Sr}$ using Sr isotope mass balance models (e.g., Kristall et al., 2017; Wang et al., 2021; Nana Yobo et al., 2021), the model approach used here calculates seawater $^{87}\text{Sr}/^{86}\text{Sr}$ over time (487 to 443 Ma) by solving a set of strontium isotopic mass balance equations (Eq. (1) and (2)) (e.g., Ingram et al., 1994):

$$\frac{dN}{dt} = J_r + J_h - J_{\text{carb}} \quad (1)$$

$$N \frac{dR_{\text{sw}}}{dt} = J_r (R_r - R_{\text{sw}}) + J_h (R_h - R_{\text{sw}}) \quad (2)$$

where t is time; N is total moles of Sr in the ocean; J_r and J_h are riverine and hydrothermal flux respectively; J_{carb} is the carbonate burial flux of Sr; R_r , R_h , and R_{sw} are riverine, hydrothermal, and seawater $^{87}\text{Sr}/^{86}\text{Sr}$, respectively. We assume a linear relationship between the Sr hydrothermal flux (J_h) (i.e., seafloor production) and long-term sea-level, which is a modified approach from the sea level-inversion technique of Gaffin (1987) and Mills et al. (2017) (see Appendix E in Supplementary Material for comparisons). The Sr hydrothermal flux (J_h) at time t is calculated using the following equation:

$$J_h(t) = \overline{J_h} \times \frac{S(t)}{\overline{S}} \quad (3)$$

where $\overline{J_h}$ is the average hydrothermal flux over the interval 487 Ma to 443 Ma (see Appendix E in Supplementary Material for further discussion), $S(t)$ is sea level in meters at time t , and $\overline{S}$ is the average sea level from 487 Ma to 443 Ma. In addition, we have held the riverine Sr flux (J_r) constant to assess the feasibility of

the hydrothermal Sr flux in driving the $^{87}\text{Sr}/^{86}\text{Sr}$ inflection. In a later section of the discussion, we allow both the hydrothermal and riverine fluxes to vary in our model approach and provide a more integrated view of the factors that drove $^{87}\text{Sr}/^{86}\text{Sr}$ to lower values through the Late Ordovician.

The results of our Ordovician Sr isotope mass balance model (Fig. 5) show that from the Tremadocian to the mid-Sandbian, the Ordovician $^{87}\text{Sr}/^{86}\text{Sr}$ curve could plausibly have been driven by the hydrothermal flux alone, including the $^{87}\text{Sr}/^{86}\text{Sr}$ inflection during the late Darriwilian. Our initial value for the hydrothermal flux is 2.0×10^{10} mol/yr (necessary to produce an early Ordovician initial value for $^{87}\text{Sr}/^{86}\text{Sr}$ of 0.7091; see Table A.6 in Supplementary Material) and reaches a maximum of 2.4×10^{10} mol/yr in the Sandbian (Fig. 5B). The peak hydrothermal flux of Sr is 50% greater than the modern hydrothermal flux of 1.6×10^{10} mol/yr (Elderfield and Schultz, 1996; Fig. 5A), which seems plausible because Ordovician seafloor production rate is estimated to be $\sim 5\text{--}6$ km²/yr, compared to the modern rate of ~ 3 km²/yr (Mills et al., 2017; Merdith et al., 2021). Jones and Jenkyns (2001) estimate an $\sim 58\%$ increase in crustal production rates by area during the Early Cretaceous (Aptian) associated with a 3×10^{-4} decrease in $^{87}\text{Sr}/^{86}\text{Sr}$, which is comparable to the 4×10^{-4} decrease in our model.

From the mid-Sandbian onward, the eustatic sea level curve changes from rising to falling (Fig. 5A) and therefore the hydrothermal flux is reduced proportionally in our model and can no longer drive the observed trend of seawater $^{87}\text{Sr}/^{86}\text{Sr}$ (Fig. 5B). An increase in the flux of Sr from continental basaltic weathering has been proposed for this time period (i.e., the Taconic Orogeny; Young et al., 2009; Swanson-Hysell and Macdonald, 2017; Conwell et al., 2022) and is necessary to drive the mid-Sandbian-Katian drop in seawater $^{87}\text{Sr}/^{86}\text{Sr}$. This increase in continental basaltic weathering flux associated with the Taconic Orogeny likely began earlier during the Sandbian (Swanson-Hysell and Macdonald, 2017) but may have been at least partly counteracted by a flux of radiogenic Sr from uplifted Precambrian basement rocks (i.e., Grenville crust) during arc-continent collision (Conwell et al., 2022). The challenge of a mass balance model such as this is the non-uniqueness of its results (e.g., Kristall et al., 2017; Nana Yobo et al., 2021; Wang et al., 2021), and the independent variables of Sr flux and $^{87}\text{Sr}/^{86}\text{Sr}$ values of those fluxes can be adjusted (along with residence time) to recreate the marine $^{87}\text{Sr}/^{86}\text{Sr}$ curve via multiple pathways. We next explore these pathways via a stochastic model.

The deterministic Sr isotope mass balance approach (e.g., Ingram et al., 1994) is valuable in demonstrating the feasibility of the hydrothermal Sr flux to act as a primary driver of the Ordovician seawater $^{87}\text{Sr}/^{86}\text{Sr}$ inflection. To complement this deterministic (forward model) approach, we have employed a second, stochastic model to reduce bias in model parameterizations (e.g., Wang et al., 2021) and demonstrate the full range of possible solutions and how likely they are. We have constructed a stochastic (inverse) model that consists of two steps: (1) Monte Carlo resampling using the steady-state assumption shown in Eq. (4) and (2) dynamic Sr modeling (see Appendix G in Supplementary Material for a more detailed description).

$$R_{\text{sw}} = \frac{J_r \times R_r + J_h \times R_h}{J_r + J_h} \quad (4)$$

Eq. (4) and the stochastic approach we employ enable us to explore a wide range of possible solutions (i.e., modeling parameters) that match the empirical Ordovician $^{87}\text{Sr}/^{86}\text{Sr}$ curve. The model produces arrays of time-dependent possible solutions using Monte Carlo subsampling from uniformly distributed parameters (Table 1). The calculations were performed 40,000 times at each

Table 1
Stochastic $^{87}\text{Sr}/^{86}\text{Sr}$ model parameters.

Symbol	Description	Initial Model Range	Source	Modern Value [*]
J_r	Riverine flux	10^9 – 10^9 mol/yr	Peucker-Ehrenbrink and Fiske (2019)	3.33×10^{10} mol/yr
R_r	Riverine isotopic composition	0.7030–0.7119	Peucker-Ehrenbrink and Fiske (2019)	0.7119
J_h	Hydrothermal flux	4.5×10^9 – 4×10^{10} mol/yr	Coogan and Dosso (2015)	1.57×10^{10} mol/yr
R_h	Hydrothermal isotopic composition	0.7030–0.7070	Antonelli et al. (2017)	0.7035

^{*} Modern values from Elderfield and Schultz (1996).

time-step. The produced arrays of solutions were then further filtered using an Ordovician $^{87}\text{Sr}/^{86}\text{Sr}$ LOESS curve with a tolerance band of $\pm 2.5 \times 10^{-5}$. The mean of the produced arrays of solutions (i.e., solution space) is then fed into a dynamic Sr model to produce the seawater $^{87}\text{Sr}/^{86}\text{Sr}$ (Fig. 6 and Supplementary Material Fig. A.9), which can be compared with the initial dynamic Sr model that was run only using changes in the hydrothermal flux (Fig. 5B and Fig. 6A). In the stochastic modeling approach the potential range for isotopic composition (R_r) of the riverine Sr flux was set to only go as high as the estimated modern value of 0.7119 (Elderfield and Schultz, 1996). We did not consider it realistic for this value to be higher during the Ordovician without the presence of continent-continent collision. The riverine flux is set to increase during the mid-Darriwilian by a maximum of 40% (Marcilly et al., 2021) due to the weatherability increase associated with the Taconic Orogeny (e.g., Conwell et al., 2022).

Fig. 6 shows the results of the stochastic $^{87}\text{Sr}/^{86}\text{Sr}$ model scenario with the riverine flux set to increase in the mid-Darriwilian (reflecting the Taconic Orogeny), thus allowing the model to predict the hydrothermal Sr flux, hydrothermal $^{87}\text{Sr}/^{86}\text{Sr}$, and riverine $^{87}\text{Sr}/^{86}\text{Sr}$ values that best produce an $^{87}\text{Sr}/^{86}\text{Sr}$ curve matching the observed seawater $^{87}\text{Sr}/^{86}\text{Sr}$ data from conodont apatite. With 700 time steps—each step representing $\sim 42,857$ years—the model produces $\sim 80,000$ sets of time-dependent solutions. This model produces an increase in the hydrothermal Sr flux during the late Darriwilian, shifting from $\sim 1.9 \times 10^{10}$ mol/yr to $\sim 2.3 \times 10^{10}$ mol/yr at 463 Ma. This is comparable to the hydrothermal flux in the sea level-driven (deterministic) model, which shifted from $\sim 2.0 \times 10^{10}$ mol/yr to $\sim 2.3 \times 10^{10}$ mol/yr (Fig. 5A). In addition, the stochastic model shows a ~ 0.001 decrease in riverine $^{87}\text{Sr}/^{86}\text{Sr}$ during the mid- to late-Darriwilian, which is supported by the geologic evidence for basalt weathering during the Taconic Orogeny (Young et al., 2009). The original Young et al. (2009) model also produces a decrease in the riverine $^{87}\text{Sr}/^{86}\text{Sr}$ of ~ 0.001 . Furthermore, Antonelli et al. (2017) show that the hydrothermal $^{87}\text{Sr}/^{86}\text{Sr}$ could also have dropped by up to 0.001 due to a shift in the relative amounts of Ca, Mg, Sr, and SO_4 in fluid entering the hydrothermal system. Although a drop in the hydrothermal $^{87}\text{Sr}/^{86}\text{Sr}$ predicted by Antonelli et al. (2017) is consistent with our stochastic model, changing the hydrothermal $^{87}\text{Sr}/^{86}\text{Sr}$ ratio alone could not produce the full magnitude of the seawater drop in $^{87}\text{Sr}/^{86}\text{Sr}$ in the Ordovician, and a decrease in riverine $^{87}\text{Sr}/^{86}\text{Sr}$ appears necessary.

Overall, the results of this stochastic (inverse) model lead us to conclude that the sea level-derived deterministic (forward) box model approach in which the hydrothermal Sr flux increases remains meaningful for the interval of time centered on the inflection point in the seawater $^{87}\text{Sr}/^{86}\text{Sr}$ curve. Later in the post-inflection time interval (beginning in the mid-Sandbian) the hydrothermal Sr flux can no longer drive the full magnitude of the $^{87}\text{Sr}/^{86}\text{Sr}$ drop, which can instead be accounted for by an increasing riverine Sr flux and decreasing riverine $^{87}\text{Sr}/^{86}\text{Sr}$ due to basaltic weathering on land (Taconic Orogeny). In addition, we note that higher sea level is typically associated with lower continental weathering rates in low lying continental interiors, which could further reduce the flux of radiogenic Sr from the craton (uplifted

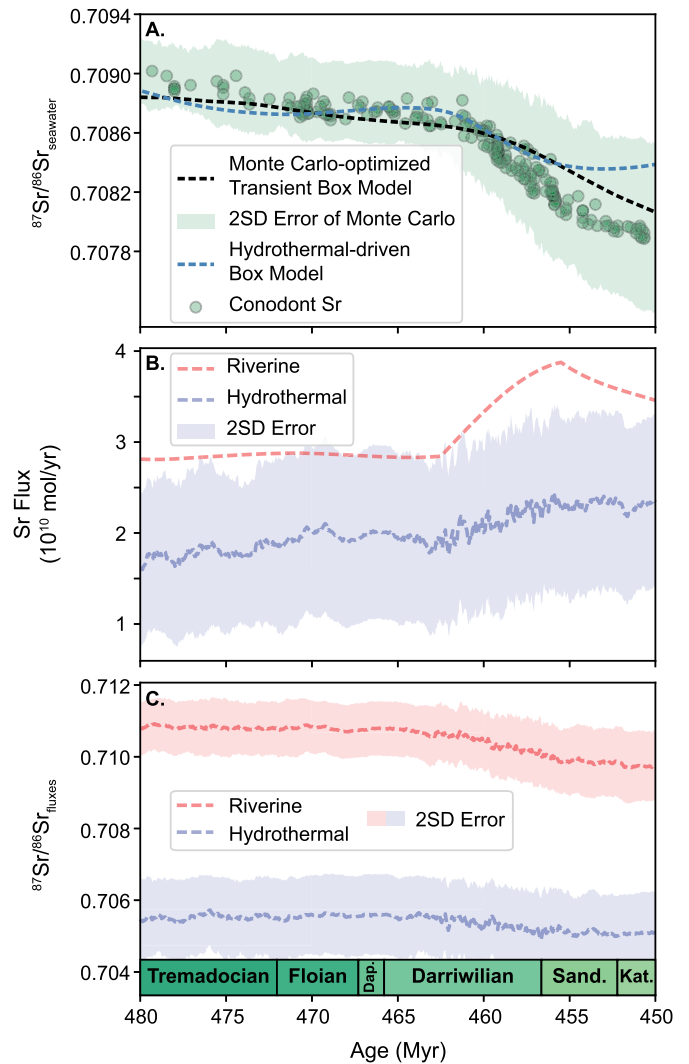


Fig. 6. Results of stochastic model of seawater Sr mass balance, produced by linking the Carbon cycle model and Monte Carlo resampling (stochastic model). For an explanation of the C cycle model see Supplementary Material Fig. A.8 and the discussion below in section 5.4. For a flow chart that explains the integration of C cycle and stochastic modeling approaches see Fig. A.9. **A)** Modeled $^{87}\text{Sr}/^{86}\text{Sr}$ based on Monte Carlo resampling compared to modeled $^{87}\text{Sr}/^{86}\text{Sr}$ driven only by hydrothermal flux from sea-level inversion (see Fig. 5B). Modeled curves are compared to the observed Ordovician seawater $^{87}\text{Sr}/^{86}\text{Sr}$ curve based on conodont data (from Saltzman et al., 2014 and this study). The hydrothermal-driven model falls within the error of the stochastic model but does not follow the observed seawater $^{87}\text{Sr}/^{86}\text{Sr}$ curve (conodont-based) for much of the Late Ordovician. **B)** Riverine Sr flux responding to 40% increase in weatherability at ~ 463 Ma following Marcilly et al. (2021) and based on evidence for weathering associated with the Taconic Orogeny (e.g., Swanson-Hysell and Macdonald, 2017; Conwell et al., 2022; and see Fig. A.8 in the supplementary material). Stochastic modeled hydrothermal Sr flux (blue dashed line) shows increase approximately matching hydrothermal flux derived from inverted sea level (see Fig. 5A). **C)** Stochastic modeled $^{87}\text{Sr}/^{86}\text{Sr}$ of riverine and hydrothermal Sr fluxes.

basalt material at a low latitude should not be reduced substantially during sea level rise).

5.4. Implications for carbon cycling and Middle to Late Ordovician cooling

Ordovician eustatic sea level changes are consistent with a model linking seafloor production rates to seawater $^{87}\text{Sr}/^{86}\text{Sr}$. Because increased seafloor production can be linked to greater subduction zone length and increased volcanic CO_2 degassing (e.g., Gaffin, 1987; Berner and Kothavala, 2001; Marcilly et al., 2021), we must consider whether our hypothesis is at odds with the apparent cooling indicated by $\delta^{18}\text{O}$ data. There are several ways to reconcile this: 1) degassing estimates using the sea level inversion technique of Gaffin (1987) are flawed because of uncertainties in the volume of ice present during the Ordovician or because seafloor spreading rates are not a good proxy for degassing (e.g., another approach is to use zircon-based ‘arc activity’ parameters of Cocks and Torsvik, 2020; see also Supplementary Material Fig. A.7); or 2) the increased CO_2 degassing rates estimated from sea level inversion are reasonable but were compensated for by rates of CO_2 consumption from the increase in Taconic basalt weathering such that cooling occurred. As a test of this second scenario, we modified the Ordovician global carbon cycle model of Young et al. (2009) and ran a series of scenarios to increase silicate weathering and therefore lower $p\text{CO}_2$. The full set of equations used in the carbon cycle model are included in the Supplementary Material (Appendix F).

To produce the observed cooling beginning in the mid-Darriwilian, we find that increased volcanic degassing must be combined with a 40% increase in weatherability (defined as the factors that drive potential silicate weathering intensity other than climate, e.g., Marcilly et al., 2021), which is greater than the 25% modeled by Young et al. (2009) (Fig. A.8 in Supplementary Material). A 40% increase in weatherability scales to a $\sim 36\%$ increase in silicate weathering on a time scale of <10 Myr, which is comparable to that observed in Phanerozoic-scale global carbon cycle models (e.g., COPSE; Mills et al., 2017). The hypothesis that the Middle to Late Ordovician may have hosted high continental weatherability is supported by studies such as Young et al. (2009), Nardin et al. (2011), Swanson-Hysell and Macdonald (2017), Marcilly et al. (2021), and Conwell et al. (2022). These studies argue for the importance of paleogeography and continental motion that increase the percentage of exposed land within 10° of the equator, in addition to the importance of uplift. Marcilly et al. (2021) demonstrate a peak in equatorial exposed land during the Middle to Late Ordovician, reaching the highest theoretical tropical weathering contribution during the Phanerozoic (23% of total exposed land). Combined with the presence of tropical orogenic belts (i.e., the Taconic Orogeny), this highlights the importance of continental weathering on Ordovician climate. Use of a spatially resolved weathering approach coupled with climatic simulations (e.g., GEOCLIM model of Nardin et al., 2011) to better quantify the impact of the Taconic orogeny on weathering rates will be necessary to provide further insights into the dynamics of Ordovician climate.

6. Conclusion

The Ordovician seawater $^{87}\text{Sr}/^{86}\text{Sr}$ inflection occurs in the late Darriwilian at c.a. $460.9 \text{ Ma} \pm 1 \text{ My}$, within the *C. friendsvillensis* and *P. serra* conodont zones. It is numerically feasible that this was driven by a surge in seafloor production that simultaneously drove a 2nd order rise in eustatic sea level, i.e., transgression associated with the Tippecanoe megasequence. Quantitative modeling demonstrates that by only changing the hydrothermal Sr flux derived from sea level (and therefore seafloor production) the

$^{87}\text{Sr}/^{86}\text{Sr}$ curve can be reproduced up to the mid-Sandbian, including the major inflection of the late Darriwilian. This suggests a greater role for hydrothermal weathering in driving seawater Ordovician $^{87}\text{Sr}/^{86}\text{Sr}$ than has been considered in previous studies. The $^{87}\text{Sr}/^{86}\text{Sr}$ inflection is accompanied by a positive shift in $\delta^{18}\text{O}$ indicating climate cooling. Modeling suggests that while a pulse in seafloor production would result in an increase in atmospheric $p\text{CO}_2$, this event combined with $\sim 36\%$ increase in continental silicate weathering can instead produce a cooling trend. It is possible that the pulse in seafloor production drove uplift that contributed to increased weatherability.

CRedit authorship contribution statement

Teresa D. Avila: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. **Matthew R. Saltzman:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing. **Y. Datu Adiatma:** Conceptualization, Formal analysis, Methodology, Software, Visualization, Writing – review & editing. **Michael M. Joachimski:** Formal analysis, Investigation, Methodology, Resources, Writing – review & editing. **Elizabeth M. Griffith:** Conceptualization, Formal analysis, Investigation, Methodology, Resources, Writing – review & editing. **John W. Olesik:** Conceptualization, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Land Acknowledgment

We have come to know the rocks and land of south-central Oklahoma in our own way, through the lens of Western science. We would like to acknowledge the people who know this land in a different and equally valuable way.

The Chickasaw Nation has known this land since about 1837, forced here from their original land—what is now known as the American Southeast—by the Indian Removal Act of 1830. A branch of the Kiikaapoi people, forced here from Mexico, has known this land since 1883. The Wichita, Waco, Tawakoni, Taovaya, and Kichai people know this land as nomads, with their forebearers traced back to the Washita River culture some 800 years ago. In 1863, the Wichita and associated tribes were forced north to what is now known as Kansas by Confederate soldiers, despite the American-Wichita treaty signed 28 years earlier in 1835 that recognized their right to their traditional homeland.

This land acknowledgment is an effort to highlight the stories of the people who have known these lands longer than we have and to celebrate their continued presence. Our work as earth scientists is focused on natural history from long before humans existed, but

that does not excuse us from the work of equity and justice in the present day. By taking time to reflect on their histories, we aim to be more attuned to modern-day indigenous groups' needs and rights.

Appendix A

Python codes related to this article can be found online at https://github.com/datuadiatma/avila_et_al_2022.

Appendix B. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.epsl.2022.117641>.

References

- Albanesi, G.L., Barnes, C.R., Trotter, J.A., Williams, I.S., Bergström, S.M., 2019. Comparative Lower-Middle Ordovician conodont oxygen isotope palaeothermometry of the Argentine Precordillera and Laurentian margins. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 549. <https://doi.org/10.1016/j.palaeo.2019.03.016>.
- Antonelli, M., Pester, N., Brown, S., DePaolo, D., 2017. Effect of paleoseawater composition on hydrothermal exchange in midocean ridges. *Proc. Natl. Acad. Sci.* 114 (47), 12413–12418. <https://doi.org/10.1073/pnas.1709145114>.
- Basu, A.R., Jacobsen, S.B., Poreda, R.J., Dowling, C.B., Aggarwal, P.K., 2001. Large groundwater strontium flux to the oceans from the Bengal Basin and the marine strontium isotope record. *Science* 293 (5534), 1470–1473. <https://doi.org/10.1126/science.1060524>.
- Bataille, C.P., Willis, A., Yang, X., Liu, X.-M., 2017. Continental crust composition: a major control of past global chemical weathering. *Sci. Adv.* 3. <https://doi.org/10.1126/sciadv.1602183>.
- Bauer, J.A., 1989. Conodont biostratigraphy and paleoecology of Middle Ordovician rocks in eastern Oklahoma. *J. Paleontol.* 63 (1), 92–107. <https://doi.org/10.1017/S0022336000041019>.
- Bauer, J.A., 2010. Conodonts and conodont biostratigraphy of the Joins and Oil Creek Formations, Arbuckle Mountains, south-central Oklahoma. *Okla. Geol. Soc. Bull.* 150.
- Berner, R.A., Kothavala, Z., 2001. GEOCARB III: a revised model of atmospheric CO₂ over Phanerozoic time. *Am. J. Sci.* 301, 182–204. <https://doi.org/10.2475/aj.s.301.2.182>.
- Carlucci, J.R., Westrop, S.R., Brett, C.E., Burkhalter, R., 2014. Facies architecture and sequence stratigraphy of the Ordovician Bromide Formation (Oklahoma): a new perspective on a mixed carbonate-siliciclastic ramp. *Facies* 60, 987–1012. <https://doi.org/10.1007/s10347-014-0412-6>.
- Cocks, L.R.M., Torsvik, T.H., 2020. Ordovician palaeogeography and climate change. *Gondwana Res.* <https://doi.org/10.1016/j.jgr.2020.09.008>.
- Conwell, C.T., Saltzman, M.R., Edwards, C.T., Griffith, E.M., Adiatma, Y.D., 2022. Nd isotopic evidence for enhanced mafic weathering leading to Ordovician cooling. *Geology*. <https://doi.org/10.1130/G49860.1>.
- Coogan, L.A., Dosso, S.E., 2015. Alteration of ocean crust provides a strong temperature dependent feedback on the geological carbon cycle and is a primary driver of the Sr-isotopic composition of seawater. *Earth Planet. Sci. Lett.* 415, 38–46. <https://doi.org/10.1016/j.epsl.2015.01.027>.
- Coogan, L.A., Gillis, K.M., 2018. Low-temperature alteration of the seafloor: impacts on ocean chemistry. *Annu. Rev. Earth Planet. Sci.* 46, 21–45. <https://doi.org/10.1146/annurev-earth-082517-010027>.
- Derby, J.R., Bauer, J.A., Creath, W.B., Dresbach, R.L., Ethington, R.L., Loch, J.D., William, M., 1991. Biostratigraphy of the Timbered Hills, Arbuckle, and Simpson Groups, Cambrian and Ordovician, Oklahoma: a review of correlation tools and techniques available to the explorationist. In: K.S. Johnson, Late Cambrian-Ordovician Geology of the Southern Midcontinent, 1989 Symposium. Oklahoma Geol. Survey, pp. 15–42.
- Edwards, C.T., Jones, C., Quinton, P., Fike, D., 2021. Oxygen isotope ($\delta^{18}\text{O}$) trends measured from Ordovician conodont apatite using secondary ion mass spectrometry (SIMS): implications for paleo-thermometry studies. *GSA Bull.* <https://doi.org/10.1130/B35891.1>.
- Edwards, C.T., Saltzman, M.R., Leslie, S.A., Bergström, S.M., Sedlacek, A.R.C., Howard, A., Young, S.A., 2015. Strontium isotope ($^{87}\text{Sr}/^{86}\text{Sr}$) stratigraphy of Ordovician bulk carbonate: implications for preservation of primary seawater values. *Geol. Soc. Am. Bull.* 127 (9–10), 1275–1289. <https://doi.org/10.1130/B31149.1>.
- Elderfield, H., Schultz, A., 1996. Mid-ocean ridge hydrothermal fluxes and the chemical composition of the ocean. *Annu. Rev. Earth Planet. Sci.* 24, 191–224. <https://doi.org/10.1146/annurev-earth.24.1.191>.
- Gaffin, S., 1987. Ridge volume dependence on seafloor generation rate and inversion using long term sea-level change. *Am. J. Sci.* 287 (6), 596–611. <https://doi.org/10.2475/aj.s.287.6.596>.
- Goldberg, S.L., Present, T.M., Finnegan, S., Bergmann, K.D., 2021. A high-resolution record of early Paleozoic climate. *Earth Planet. Sci. Lett.* <https://doi.org/10.1016/j.pnas.2013083118/-/DCSupplemental>.
- Goldman, D., Sadler, P., Leslie, S.A., 2020. The Ordovician period. In: Gradstein, F.M., Ogg, J.G., Schmitz, M.D., Ogg, G.M. (Eds.), *Geologic Time Scale 2020*, pp. 631–694.
- Grossman, E.L., Joachimski, M.M., 2020. Oxygen isotope stratigraphy. In: Gradstein, F.M., Ogg, J.G., Schmitz, M.D., Ogg, G.M. (Eds.), *Geologic Time Scale 2020*, pp. 279–307.
- Haq, B.U., Schutter, S.R., 2008. A chronology of Paleozoic sea-level changes. *Science* 322, 64–68. <https://doi.org/10.1126/science.1161648>.
- Holmden, C., Creaser, R.A., Muehlenbachs, K., Bergström, S.M., Leslie, S.A., 1996. Isotopic and elemental systematics of Sr and Nd in 454 Ma biogenic apatites: implications for paleoseawater studies. *Earth Planet. Sci. Lett.* 142, 425–437.
- Ingram, B.L., Coccioni, R., Montanari, A., Richter, F.M., 1994. Strontium isotopic composition of mid-Cretaceous seawater. *Science* 264, 546–550. <https://doi.org/10.1126/science.264.5158.546>.
- Joachimski, M.M., Breisig, S., Buggisch, W., Talent, J.A., Mawson, R., Gereke, M., et al., 2009. Devonian climate and reef evolution: insights from oxygen isotopes in apatite. *Earth Planet. Sci. Lett.* 284, 599–609. <https://doi.org/10.1016/j.epsl.2009.05.028>.
- Jones, C.E., Jenkyns, H.C., 2001. Seawater strontium isotopes, oceanic anoxic events, and seafloor hydrothermal activity in the Jurassic and Cretaceous. *Am. J. Sci.* 301 (2), 112–149. <https://doi.org/10.2475/aj.s.301.2.112>.
- Kristall, B., Jacobson, A.D., Hurtgen, M.T., 2017. Modeling the paleo-seawater radiogenic strontium isotope record: a case study of the Late Jurassic-Early Cretaceous. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 472, 163–176. <https://doi.org/10.1016/j.palaeo.2017.01.048>.
- Lenton, T.M., Crouch, M., Johnson, M., Pires, N., Dolan, L., 2012. First plants cooled the Ordovician. *Nat. Geosci.* 5, 86–89. <https://doi.org/10.1038/ngeo1390>.
- Macdonald, F.A., Swanson-Hysell, N.L., Park, Y., Lisiecki, L., Jagoutz, O., 2019. Arc-continent collisions in the tropics set Earth's climate state. *Science* 364, 181–184. <https://doi.org/10.1126/science.aav5300>.
- Männik, P., Lehnert, O., Nölvak, J., Joachimski, M.M., 2021. Climate changes in the pre-Hirnantian Late Ordovician based on $\delta^{18}\text{O}_{\text{phos}}$ studies from Estonia. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 569, 110347. <https://doi.org/10.1016/j.palaeo.2021.110347>.
- Marcilly, C.M., Torsvik, T.H., Domeier, M., Royer, D.L., 2021. New paleogeographic and degassing parameters for long-term carbon cycle models. *Gondwana Res.* 97, 176–203. <https://doi.org/10.1016/j.jgr.2021.05.016>.
- McArthur, J.M., Howarth, R., Shields, G.A., Zhou, Y., 2020. Strontium isotope stratigraphy. In: Gradstein, F., Ogg, J.G., Schmitz, M., Ogg, G. (Eds.), *Geologic Time Scale 2020*, pp. 211–238.
- McKenzie, R.N., Horton, B.K., Loomis, S.M., Stockli, D.F., Planavsky, N.J., Lee, C.-T.A., 2016. Continental arc volcanism as the principal driver of icehouse-greenhouse variability. *Science* 352 (6284), 444–447. <https://doi.org/10.1126/science.aad5787>.
- Merdith, A.S., Collins, A.S., Williams, S.E., Pisarevsky, S., Foden, J.D., Archibald, D.B., et al., 2021. A full-plate global reconstruction of the Neoproterozoic. *Earth-Sci. Rev.* 214, 103477. <https://doi.org/10.1016/j.jgr.2017.04.001>.
- Mills, B.J.W., Scotese, C.R., Walding, N.G., Shields, G.A., Lenton, T.M., 2017. Elevated CO₂ degassing rates prevented the return of Snowball Earth during the Phanerozoic. *Nat. Commun.* 8 (1), 1–7. <https://doi.org/10.1038/s41467-017-01456-w>.
- Müller, D., Dutkiewicz, A., 2018. Oceanic crustal carbon cycle drives 26-million-year atmospheric carbon dioxide periodicities. *Sci. Adv.* 4 (2). <https://doi.org/10.1126/sciadv.aag0500>.
- Nana Yobo, L., Brandon, A.D., Holmden, C., Lau, K., Eldrett, J., 2021. Changing inputs of continental and submarine weathering sources of Sr to the oceans during OAE 2. *Geochim. Cosmochim. Acta* 303, 205–222. <https://doi.org/10.1016/j.gca.2021.03.013>.
- Nardin, E., Goddard, Y., Donnadieu, Y., le Hir, G., Blakey, R.C., Pucéat, E., Aretz, M., 2011. Modeling the early Paleozoic long-term climatic trend. *Bull. Geol. Soc. Am.* 123 (5), 1181–1192. <https://doi.org/10.1130/B30364.1>.
- Peucker-Ehrenbrink, B., Fiske, G.J., 2019. A continental perspective of the seawater $^{87}\text{Sr}/^{86}\text{Sr}$ record: a review. *Chem. Geol.* 510, 140–165. <https://doi.org/10.1016/j.chemgeo.2019.01.017>.
- Rasmussen, C.M., Ullmann, C.V., Jakobsen, K.G., Lindskog, A., Hansen, J., Hansen, T., Harper, D.A.T., 2016. Onset of main Phanerozoic marine radiation sparked by emerging Mid Ordovician icehouse. *Sci. Rep.* 6, 1–9. <https://doi.org/10.1038/srep18884>.
- Saltzman, M.R., Edwards, C.T., Leslie, S.A., Dwyer, G.S., Bauer, J.A., Repetski, J.E., Bergström, S.M., 2014. Calibration of a conodont apatite-based Ordovician $^{87}\text{Sr}/^{86}\text{Sr}$ curve to biostratigraphy and geochronology: implications for stratigraphic resolution. *Bull. Geol. Soc. Am.* 126 (11–12), 1551–1568. <https://doi.org/10.1130/B31038.1>.
- Scher, H.D., Griffith, E.M., Buckley, W.P., 2014. Accuracy and precision of $^{88}\text{Sr}/^{86}\text{Sr}$ and $^{87}\text{Sr}/^{86}\text{Sr}$ measurements by MC-ICPMS compromised by high barium concentrations. *Geochim. Geophys. Geosyst.* 15 (2), 499–508. <https://doi.org/10.1002/2013GC005134>.
- Scotese, C.R., Wright, N.M., 2018. PALEOMAP paleodigital elevation models (PaleoDEM) for the Phanerozoic. <https://doi.org/10.5281/zenodo.5460860>.

- Shields, G.A., Carden, G.A.F., Veizer, J., Meidla, T., Rong, J.Y., Li, R.Y., 2003. Sr, C, and O isotope geochemistry of Ordovician brachiopods: a major isotopic event around the Middle-Late Ordovician transition. *Geochim. Cosmochim. Acta* 67 (11), 2005–2025. [https://doi.org/10.1016/S0016-7037\(02\)01116-X](https://doi.org/10.1016/S0016-7037(02)01116-X).
- Steiger, R.H., Jäger, E., 1977. Subcommission on geochronology: convention on the use of decay constants in geo- and cosmochronology. *Earth Planet. Sci. Lett.* 36, 359–362.
- Swanson-Hysell, N.L., Macdonald, F.A., 2017. Tropical weathering of the Taconic orogeny as a driver for Ordovician cooling. *Geology* 45 (8), 719–722. <https://doi.org/10.1130/G38985.1>.
- Trotter, J.A., Williams, I.S., Barnes, C.R., Lecuyer, C., Nicoll, R.S., 2008. Did cooling oceans trigger Ordovician biodiversification? Evidence from conodont thermometry. *Science* 321, 550–554. <https://doi.org/10.1126/science.1158908>.
- Van Der Meer, D., Zeebe, R., Van Hinsbergen, D., Sluijs, A., Spakman, W., Torsvik, T., 2014. Plate tectonic controls on atmospheric CO₂ levels since the Triassic. *Proc. Natl. Acad. Sci.* 111 (12), 4380–4385. <https://doi.org/10.1073/pnas.1315657111>.
- Vennemann, T.W., Fricke, H.C., Blake, R.E., O'Neil, J.R., Colman, A., 2002. Oxygen isotope analysis of phosphates: a comparison of techniques for analysis of Ag₃PO₄. *Chem. Geol.* 185, 321–336.
- Wang, W., Katchinoff, J.A.R., Garbelli, C., Immenhauser, A., Zheng, Q., Zhang, Y., Yuan, D., Shi, Y., Wang, J., Planavsky, N., Shen, S., 2021. Revisiting the Permian seawater ⁸⁷Sr/⁸⁶Sr record: new perspectives from brachiopod proxy data and stochastic oceanic box models. *Earth-Sci. Rev.* 218. <https://doi.org/10.1016/j.earscirev.2021.103679>.
- Young, S.A., Saltzman, M.R., Foland, K.A., Linder, J.S., Kump, L.R., 2009. A major drop in seawater ⁸⁷Sr/⁸⁶Sr during the Middle Ordovician (Darriwilian): links to volcanism and climate? *Geology* 37 (10), 951–954. <https://doi.org/10.1130/G30152A.1>.